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A3-PS036-TC187

AIORI HACKATHON 3

PQC DNSSEC MTL benchmark

A measured comparison of SLH-DSA with and without Merkle Tree Ladder batching

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MENTOR-APPROVED BENCHMARK SCOPE | AIORI-3 / PS-036

Problem Description and Abstract

Problem description

- SLH-DSA signatures are large for DNS answers; the measured plain signature is 7,856 bytes.
- MTL amortizes one expensive signature over a batch and supplies shorter per-record proofs.
- The experiment compares plain signing, MTL batch one, and full-batch MTL on the same synthetic RRsets.
- Page 1 of PS-036 is reviewed; pinned Docker build and checks passed on the student's host.

Abstract

LadderLens measures the reference implementation rather than proposing new cryptography. The retained native WSL2 run recorded 216 successful trials and 15 cache replays. At 1,000 RRsets, the measured full-batch MTL median signing total was 929.3 ms; the plain total was 944.85 s extrapolated from 100 sampled records. The student reports mentor approval of the current methodology. The pinned Docker image build and requested checks passed on the student's host; the retained performance run itself was not rerun in Docker. Page 1 of PS-036 is reviewed; page 2 is excluded from this review at the student's direction.

Background

Why MTL matters

Post-quantum DNSSEC has a bandwidth and computation trade-off. MTL signs a Merkle tree ladder once, then authenticates each RRset with a compact authentication path. A resolver can reuse a checked ladder for later answers. The project measures this workflow using the Verisign MTL reference library and liboqs.

Plain signature

7,856 B measured per SLH-DSA signature. This is signature material, not a complete DNS packet.

Signed ladder

7,960 B at N=10; 8,088 B at N=1,000. A resolver fetches and checks a ladder before warm verification.

Condensed proof

Largest measured proof: 124 B at N=10 to 220 B at N=1,000, before the one-byte MTL type field.

DNSSEC signing input is constructed from canonical RRsets and RRSIG fields. The fixture uses a synthetic mix of A, AAAA, TXT and MX records; it is not a production-zone trace.

Literature & RFC Review

FIPS 205 - SLH-DSA

Defines the underlying stateless hash-based signature scheme used for the plain baseline and MTL ladder signatures.

RFC 4034, Sections 3.1.8.1 and 6

Defines RRSIG signing data and canonical RRset form used to construct fixture messages.

I-D draft-harvey-cfrg-mtl-mode

Describes full/condensed MTL signatures, ladders and authentication paths. It remains an Internet-Draft, not a standard.

I-D draft-fregly-dnsop-slh-dsa-mtl-dnssec-06

Describes the DNSSEC application, MTL type prefix and retrieval behavior relevant to the benchmark.

RFC 9715

Provides EDNS(0) context for DNS UDP payload sizing; LadderLens constructs local messages and does not capture packets.

The experiment uses these documents as technical references, not as organizer approval. Student-reported mentor approval of the current choices is noted; the written message is not retained.

Proposed Solution

A - Plain SLH-DSA

One zone key; one independent signature and verification per RRset.

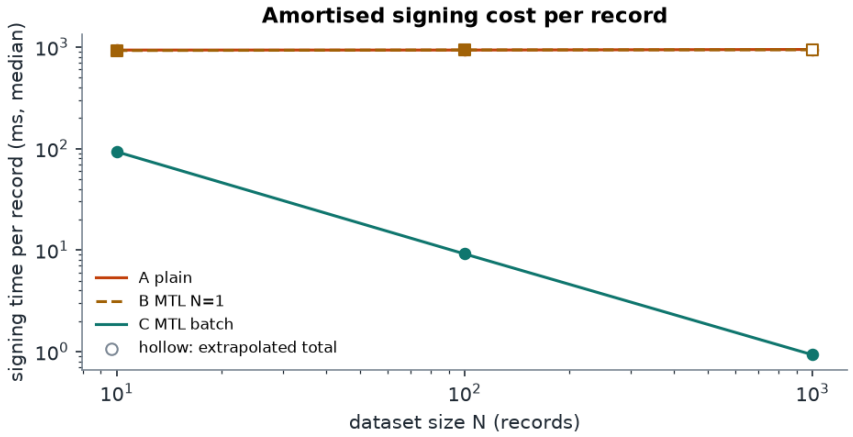
B - MTL batch one

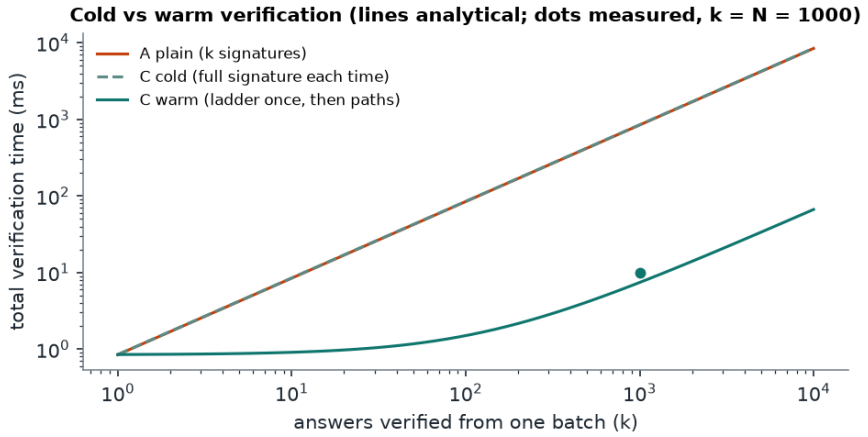
One MTL tree/key per RRset; full signature. Diagnostic arm for no-sharing overhead.

C - Full-batch MTL

One key and signed ladder for N RRsets; one condensed proof per RRset.

The C harness calls the library's public API. Python creates deterministic DNSSEC fixtures, plans interleaved trials, verifies manifests, analyzes results and generates chart

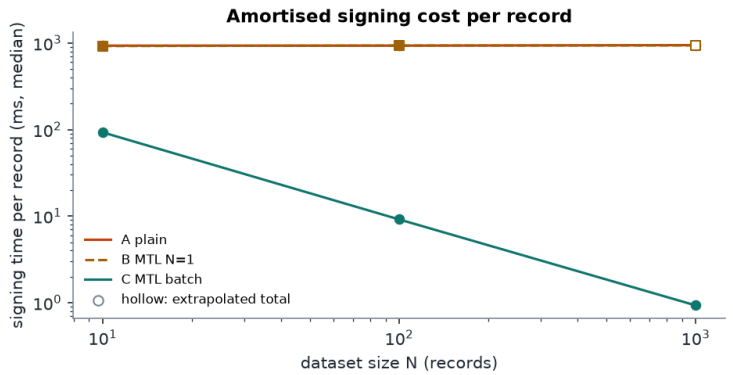




Optimization Proposed - Measured Results

Median full-dataset signing time at N=1,000

Arm	Records timed per trial	Median signing total	Evidence tag
Plain SLH-DSA	100 of 1,000	**944.85 s**	Extrapolated
MTL batch one	100 of 1,000	**937.84 s**	Extrapolated
Full-batch MTL	1,000 of 1,000	**929.3 ms**	Measured

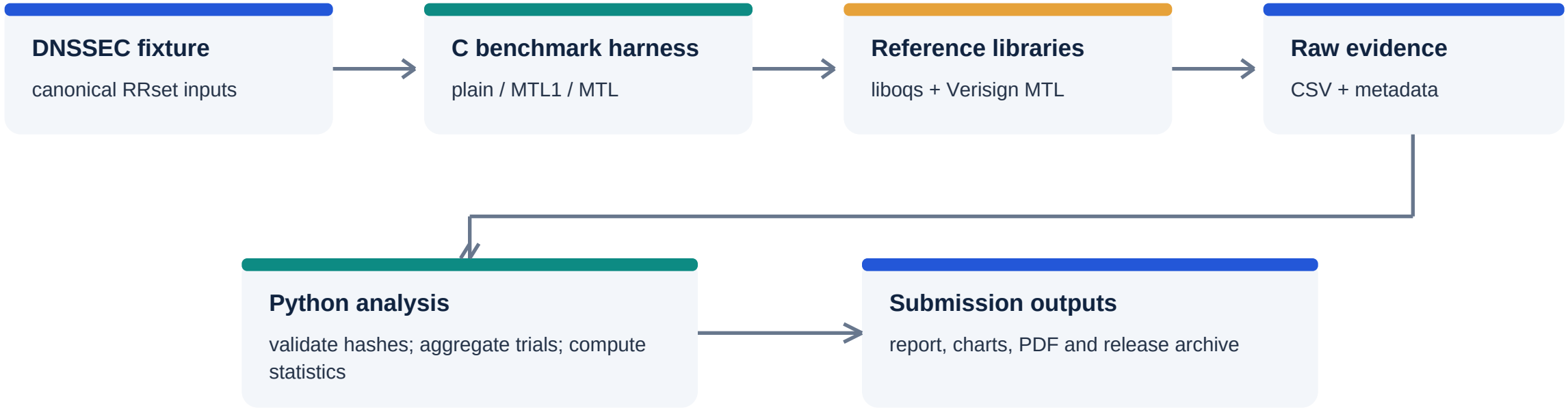


The retained run exceeded its 120-minute planning target: 163.5 minutes. At N=1,000, the first two totals scale evenly spaced 100-record trials; they are not direct 1,000-record measurements. Trial confidence intervals do not include RRset-sampling uncertainty.

Bytes are separate: signature material is measured; complete DNS response lengths are locally encoded analytical values; actual network traffic is not measured. Cache replay is a synthetic workload, not production traffic.

Solution Architecture & Design

Measurement architecture - implementation uses reference-library APIs; no new cryptographic primitive is implemented.



Separate cache-replay executable exercises real MTL verification calls and records hits, misses and bytes for a synthetic Zipf query stream.

Data Flow Diagram

1	**1. Generate**	Seeded zone: N = 10, 100, 1,000 RRsets; same input across arms.
2	**2. Plan and sample**	Pilot determines trial count. Plain and batch-one may process K < N records.
3	**3. Measure**	Interleaved harness calls record per-operation timings and signature sizes.
4	**4. Preserve**	Run manifest records configuration, environment, hashes and invocation schedule.
5	**5. Analyze**	Validate exact files and hashes; summarize trials; tag measured, extrapolated and analytical values.
6	**6. Report**	Tables and charts; constructed DNS response lengths kept distinct from wire bytes.

Deployment Timeline and Methodology

Done - source and retained run

216/216 trials and 15/15 cache replays. Native WSL2 replica; not Docker.

Done - methodology approval reported

Current sizes, separated byte reporting, and N=1,000 extrapolation approved; written message not retained.

Pinned Docker checks passed

Image built; harness compiled; smoke passed; captured verify output passes all checks; 20/20 Python tests passed, including end-to-end.

Page 1 technical demonstration

Prototype demo using the benchmark and pseudocode; no presentation evidence is retained in this workspace.

Scope of this review

Page 1 technical brief assessed; page 2 is excluded at the student's direction.

Bibliography

[1] NIST, FIPS 205, Stateless Hash-Based Digital Signature Standard, August 2024. https://doi.org/10.6028/NIST.FIPS.205

[2] R. Arends et al., RFC 4034, Resource Records for the DNS Security Extensions, March 2005. https://www.rfc-editor.org/rfc/rfc4034

[3] J. Harvey et al., draft-harvey-cfrg-mtl-mode-09, Merkle Tree Ladder (MTL) Mode Signatures, March 2026. Internet-Draft; work in progress.

[4] A. Fregly et al., draft-fregly-dnsop-slh-dsa-mtl-dnssec-06, SLH-DSA-MTL for DNSSEC, March 2026. Internet-Draft; work in progress.

[5] IETF RFC 9715, IP Fragmentation Avoidance in DNS over UDP, May 2025.

[6] Verisign MTL reference implementation, commit 7cef0191da55048111d252a567b7614672424ed3; the image build checks the commit before compiling.

[7] Open Quantum Safe liboqs 0.16.0, commit 5a1a854b0dc9f2141bdc771c555ee60c37950183; OpenSSL 3.6.0, commit 7b371d80d959ec9ab4139d09d78e83c090de9779.

[8] AIORI-3 Hackathon Guidelines and Proposed Structure PDF, public organizer materials. https://portal.aiori.in/aiori-3/hackathon-guideline/

[9] AIORI-3 PS-036, PQC DNSSEC MTL Benchmark Using a Reference Implementation, two-page assignment supplied by the team; preserved in docs/reference/PS-036.pdf.

The student reports mentor approval of current methodology choices; the written response is not retained. See docs/PS036-COMPLIANCE.md for requirement evidence and limitations.

LL

Thank You

LadderLens v6 - AIORI-3 PS-036

Current status: Docker checks passed; demo evidence not retained

Next: if not already completed, present the page 1 technical demonstration using the prototype and pseudocode. The retained performance benchmark remains a native WSL2 run, not a Docker rerun. Page 2 is outside this review at the student's direction.

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Docker validation checks passed; remote CI and formal approval are not asserted.